

Clinical and Pathophysiological Overview of Spinal Muscular Atrophy Type 1 and Type 2

Genomic Basis, Splicing Mechanics, Molecular Motor Unit Degeneration, and Clinical Phenotypes

A. Genomic Architecture & Splicing Mechanics (SMN1 vs. SMN2)

Homozygous SMN1 Loss / Deletion (Telomeric SMN1)

Exon 6 Intron 6 Exon 7 (C at c.840) Exon 8

c.840C creates ESE heptamer binding SF2/ASF -> 100% Full-Length SMN Transcript

Result: Stable, oligomerization-competent ~38 kDa Functional SMN Protein

Centromeric SMN2 Gene Modifier (Centromeric SMN2)

Exon 6 Intron 6 Exon 7 (T at c.840) Exon 8

c.840C>T transition disrupts ESE and creates ESS -> 85-90% Exon 7 Skipping

Result: 85-90% Truncated SMN-delta7 (rapidly degraded via ubiquitin-proteasome); 10-15% Full-Length SMN

B. Neuromuscular Unit Pathology & Agrin-MuSK Denervation

1. Spinal Motor Neuron Perikaryon Degeneration:

Loss of SMN disrupts snRNP biogenesis and pre-mRNA splicing in anterior horn alpha-motor neurons.

2. Impaired Axonal Transport & Neuromuscular Junction Synaptogenesis:

Disrupted Agrin/MuSK/LRP4 signaling prevents acetylcholine receptor clustering at motor endplates.

3. Progressive Retrograde Denervation & Myofiber Atrophy:

Terminal axon retraction, muscle denervation, and secondary grouped neurogenic atrophy.

4. Electrophysiological Correlate:

Marked reduction in Compound Muscle Action Potential (CMAP) amplitude (Type 1: <1.0 mV; Type 2: 1.5-3.5 mV).

C. Phenotypic Stratification: SMA Type 1 vs. SMA Type 2

SMA Type 1 (Werdnig-Hoffmann Disease) - Severe Non-Sitter

Onset: 0 to 6 months (mean: 2.4 months) | SMN2 copies: Typically 2 (rarely 3)

Milestones: Never achieve independent sitting; severe generalized hypotonia & head lag

Natural History: Median survival <2 years without mechanical ventilatory support

SMA Type 2 (Dubowitz Disease) - Intermediate Sitter

Onset: 7 to 18 months (mean: 11.1 months) | SMN2 copies: 3 to 4 copies

Milestones: Achieve independent sitting unsupported; never walk independently

Complications: Progressive scoliosis, joint contractures, restrictive pulmonary decline

D. Standardized Motor Scales & Therapeutic Intervention Timeline

1. Standardized Psychometric Motor Scales:

CHOP-INTEND (Type 1): 16 items, 0-64 pts (Head/trunk control, reach, kicking). MCID >= 4 pts.

HFMSE (Type 2): 33 items, 0-66 pts (Sitting, rolling, crawling, standing, squatting). MCID >= 3 pts.

2. Disease-Modifying Therapies (DMTs):

Nusinersen: Intrathecal ASO targeting ISS-N1 to promote SMN2 exon 7 inclusion.

Risdiplam: Daily oral small-molecule SMN2 pre-mRNA splicing modifier.

Onasemnogene Apeparovvec: AAV9 intravenous gene replacement supplying functional SMN1 cDNA.

Crucial Prognostic Window: Early treatment prior to irreversible motor unit loss dictates long-term response.